NIGERIA · 183 CITIES · 1990 TO 2020

A centre that holds less

The share of people at the heart of Nigeria's cities, 1990 to 2020

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ABSTRACT

In Mexico the centres of cities lost 2.5 million residents between 1990 and 2020 while the edges filled. This paper asks whether Nigeria's cities did the same. It is an adaptation of Peraza-Mues and colleagues' study of Mexican urban centres, run on the 183 Nigerian urban centres that already existed in 1990, each held inside its 2025 GHSL outline, with people from the GHS-POP grid at 100 metres. Each city gets one centre, 100 metre rings around it, and four zones scaled to its size. The share of urban Nigerians living in a central zone fell from **39.1%** to **29.9%**, and it fell in **177 of 183** cities, by a median of **13 points**. The exception is size: in the fourteen cities of over a million the median fall was **3.7 points**, and Lagos moved by one. In absolute terms the central zones gained **9.7 million** people, which reads like the opposite of Mexico. It is not evidence of that. Run through the same steps, the same grid turns Mexico's census loss of 2.5 million into a gain of **1.9 million**. A second grid, WorldPop, barely changes shape from year to year. What the satellite record can carry in Nigeria is the share. The count at the centre needs a census.

KEYWORDS urban expansion; city centres; GHS-POP; WorldPop; remoteness; Lagos; Kano; Ibadan; Port Harcourt; Abuja; Mexico

Glossary

Urban centre

A cluster of contiguous 1 km cells with at least 1,500 people per square kilometre and 50,000 people in all, as drawn by the Global Human Settlement Layer (GHSL) for 2025. Nigeria has 425. Each outline is held fixed for every year in this paper.

Panel

The 183 urban centres whose GHSL year of birth is 1990 or earlier. Only these enter the 1990 to 2020 comparison. A city that was a village in 1990 cannot lose a centre it did not have.

Centre

One point per city. For Lagos, Kano, Ibadan and Port Harcourt, a founding place. For Abuja, the Three Arms Zone, its planned origin. Everywhere else, the densest point of 1975 built-up surface, smoothed over about a kilometre.

Ring

A band 100 metres wide around the centre. People are counted ring by ring, with distance measured on the ground.

Remoteness, r

Distance from the centre in kilometres, times 1,000, divided by the square root of the city's 2020 population. It lets a ring in Lagos and a ring in Wukari be compared. A bigger city has a wider central zone.

Zones

Central (r below 3), intermediate (3 to 5), distant (5 to 9.3) and peri-urban (above 9.3), the anchor paper's cut-offs. Lagos's central zone reaches 10.3 km from Lagos Island; Abuja's reaches 3.2 km.

Central share

People in the central zone divided by people in the whole outline, in the same year. A change in the share is reported in percentage points (points).

Expansion factor, Φ

How much farther from the centre the typical resident lives in a later year. It is the slope of one year's distance percentiles against another's. A Φ of 1.10 means distances stretched by a tenth.

β and α

The two numbers of the anchor's model of Φ. β says how much the stretch grows with population growth. α is the stretch per year that growth does not explain.

Distance ratio

How far people live from the centre after thirty years, against a city that grew at constant density. Below 1 means closer than that city; above 1, farther.

GHS-POP

The GHSL population grid, release R2023A, at 100 metres. It takes census totals for administrative areas and spreads them over built-up volume seen by satellite. It is a model, not a count.

WorldPop

A second population grid at about 100 metres, made for each year from 2000 to 2020 with a random forest on roads, lights, land cover and settlement maps, adjusted to United Nations totals.

Every share in the results is a share of people, not of land. All numbers come from the 183-city panel unless a city is named.

1. Introduction

The centre of a city is where it began, and for a long time it is where most of its people lived. Then the edge became cheaper, the roads went out, the old houses became shops, and the centre stopped holding the people it used to hold. That is the story of the American city after the war, of many European cores, and, according to a careful recent study, of Mexico.

Peraza-Mues, Resendiz, Figueroa-Soriano, Prieto-Curiel and Ponce-Lopez (2025) took the Mexican census at block level for 1990, 2000, 2010 and 2020, drew rings 100 metres wide around the historic founding point of 69 cities, and counted. The central zones lost 2.5 million residents in thirty years. The share of urban Mexicans living centrally fell from over 40% to 22%. Their model of how cities stretch, $\ln \Phi = \beta \ln(\text{growth}) + \alpha \Delta t$, found $\beta = 0.60$ and $\alpha = 0.0057$: a Mexican city that doubled spread its people out by more than its growth alone required, and distances to the centre ended 28% longer than in a city that had grown at constant density.

This paper is an adaptation of that study to Nigeria. Nigeria is a harder place to ask the question. Its cities grew faster than Mexico's, its last census was in 2006, and the one before was in 1991. There is no block-level count for 1990, 2000, 2010 and 2020. What exists is a set of global population grids that start from census totals and spread people over what satellites see. So the paper asks two things at once. Did the centre of Nigerian cities hold less of their people in 2020 than in 1990? And which parts of that answer can a satellite-based grid be trusted to give?

2. Statement of the problem

The sentence that fails in Nigeria is "the centre is emptying". It might be true. Lagos Island has become a place of offices and markets, and the old walled city of Kano has been rebuilt shop by shop. But nobody has counted the people in those places on the same ground in 1990 and 2020, and a grid that puts people where buildings are cannot see a house become a warehouse. It will keep the people in the building.

That means the measurement problem is the finding, not a footnote to it. If the anchor's method is run on GHS-POP and the centre gains people, the gain could be the cities or it could be the grid. A comparison is needed that separates the two: the same steps, on the same grid, in a country where the census already says what happened.

The question this paper can answer is therefore narrower than "did Nigerian centres empty":

1. How did the share of people in each remoteness zone change between 1990 and 2020, in the 183 cities that existed in 1990?
2. Does that change differ by city size, and in the five cities people would ask about first: Lagos, Kano, Ibadan, Port Harcourt and Abuja?
3. When the same steps are run on Mexico, does GHS-POP reproduce the census loss, and does a second grid agree with the first?

3. Justification

Why this study. Where people live relative to the centre decides the length of the commute, the reach of the old markets and the pressure on the edge, where land is cheap and services are thin. Nigerian planning documents often assume either a crowded core or an exploding fringe. Neither has been measured on a common footing across the country's cities.

Why these places. All 425 GHSL urban centres in Nigeria are in scope, because the question is national, but the comparison over thirty years uses the 183 that were urban centres by 1990. Five are named because readers will look for them: Lagos, the largest; Kano, the old northern capital with a walled core; Ibadan, the old Yoruba city that spread across its hills; Port Harcourt, a colonial port on the creeks; and Abuja, planned from 1976 and built from the 1980s, which has an origin but no history at its centre.

Why this design. The anchor's method needs one centre per city, a fixed outline, and the same population source in every year. GHS-POP is the only grid that covers 1990 to 2020 at 100 metres on one method. WorldPop starts in 2000 and is used as a check. Mexico is run through the same pipeline

because the anchor's census numbers are the only yardstick available for how GHS-POP behaves at a city's centre.

What it is entitled to claim. A fall in the central share, if it survives the Mexican test, is a defensible reading of where growth went. A gain or loss of people at the centre, in absolute terms, is not, unless the Mexican test says GHS-POP can see such a loss. It cannot, as Section 6.4 shows, so this paper does not claim one.

4. Study area

Nigeria's 425 urban centres are spread across all 36 states and the Federal Capital Territory (Figure 1, on the landscape page after Section 5). The 183 that existed by 1990 held 39.4 million people inside their 2025 outlines in 1990 and 84.1 million in 2020, by GHS-POP. The other 242 became urban centres later, many of them on the edge of Lagos, in the Abuja corridor, and in the dense south-east around Onitsha, Aba and Owerri.

The outline is GHSL's 2025 urban centre, not a state, a local government area or a metropolitan plan. It is fixed for every year, so a ring that was farmland in 1990 and housing in 2020 is in both counts. That is the anchor's choice too: the 2020 central urban cluster of each Mexican metropolitan area, held fixed.

Lagos is the 2025 outline from Lagos Island across the mainland, east along the Lekki peninsula and north across the state line into Ogun, 11.7 million people in 2020. Ikorodu is its own urban centre. Kano is the walled city and Fagge, Sabon Gari, Nassarawa and the ring of newer districts, 4.5 million. Ibadan is the old city around Mapo Hill and its spread to Bodija, Akobo and Apete, 3.5 million. Port Harcourt is the island Township, Diobu and the mainland north to Choba and Rumuokoro, 2.1 million. Abuja is the Federal Capital City from Asokoro and Maitama west to Gwarinpa and Kubwa, 1.1 million. Abuja's GHSL birth year is 1995, so it sits outside the panel.

5. Data and methods

Figure 2, on the landscape page after Figure 1 at the end of this section, is the whole method on one plate. Read it left to right. Three sources make one centre and one set of 100 metre rings for every city. The rings make the zones, the expansion factor and the model. The same steps run on Mexico.

5.1 Outlines and people

Outlines are the GHSL Urban Centre Database, release R2024A version 1.2 (European Commission, 2024), sub-Saharan Africa file, kept for Nigeria. The fields used are the identifier, the main name, the 2025 population and the year of birth.

People are GHS-POP R2023A (Schiavina et al., 2023) for the epochs 1990, 2000, 2010 and 2020, read from Google Earth Engine on the grid's native 100 metre Mollweide cells. Built-up surface in 1975 is GHS-BUILT-S R2023A (Pesaresi and Politis, 2023). WorldPop's unconstrained 100 metre grids for Nigeria in 2000, 2010 and 2020 (Tatem, 2017) are resampled to the same cells as a density, so that each 100 metre square receives its share of the people in the coarser WorldPop cell. Each city is read with a 2 kilometre margin and then cut to its outline.

5.2 One centre per city

The anchor uses the historic founding point, usually the main square. Four of the named cities have one: Iga Idunganran, the Oba's palace on Lagos Island; Gidan Rumfa, the Emir's palace in Kano; Mapo Hill in Ibadan; and the old Township waterfront in Port Harcourt. Abuja's is the Three Arms Zone, where the plan begins. For the other 420 there is no list of founding points. The centre is the densest place of 1975 built-up surface, after a 1.1 kilometre moving average. That is the oldest settled ground the satellite record has. It is not the 1990 population peak, which would put the centre wherever people already were in the first year and bias the result toward a loss.

Each centre was checked on a street map, for the five named cities and for the cities whose centre lost the most people. All sit on their old towns. Port Harcourt's point lies about 2 kilometres south of the 1975 peak, on the waterfront, so part of its central zone is creek. Abuja's lies outside GHSL's own outline of the city, which explains its tiny central zone in Section 6.2.

5.3 Rings and zones

Each 100 metre cell is split into 25 points, 20 metres apart, each carrying a 25th of the cell's people. The distance from each point to the centre is measured on the ground, in an azimuthal equidistant projection centred on that city. The GHSL grid itself keeps area but not distance: near Nigeria it squeezes east to west and stretches north to south by about a tenth. People are then counted in 100 metre rings.

$$\text{remoteness } r = \frac{1000 \times \text{distance (km)}}{\sqrt{\text{people in 2020}}} \quad (1)$$

$$\text{density } \sigma(s) = \frac{\text{people in the ring at } s}{\text{area of the ring}} \quad (2)$$

Central $r < 3$, intermediate 3 to 5, distant 5 to 9.3, peri-urban above 9.3, as in the anchor.

> Kano had 4,452,007 people inside its outline in 2020. The square root is 2,110. A remoteness of 3 is $3 \times 2{,}110 / 1{,}000 = 6.3$ kilometres from Gidan Rumfa. Everyone within 6.3 kilometres is central, in every year. Wukari, with 165,907 people, has a central zone of 1.2 kilometres. The zones scale with the city, so that a market town and a megacity are compared on the same terms.

The ring area includes water and empty land, as in the anchor. Lagos's central zone is about half lagoon and sea, which lowers its density but not its change.

5.4 Expansion factor and model

For each city and each year the radial distances below which 1%, 2%, up to 99% of people live are read off the rings. Φ for a pair of years is the slope of the later year's percentiles against the earlier year's, a line through the origin. With four years there are six pairs per city. The anchor's model is fitted to all of them by ordinary least squares, and its uncertainty by resampling whole cities 2,000 times.

$$\ln \Phi = \beta \ln(P_{later} / P_{earlier}) + \alpha (t_{later} - t_{earlier}) \quad (3)$$

$$\text{distance ratio} = \text{growth}^{\beta - 0.5} \cdot e^{\alpha \Delta t} \quad (4)$$

A city that grows at constant density stretches by the square root of its growth, so $\beta = 0.5$ and $\alpha = 0$ give a ratio of 1.

5.5 Two tests

The first test is Mexico. The 182 GHSL urban centres of Mexico, 92 of them urban by 1990, are put through the same download, the same rings and the same zones, with GHS-POP. Mexico City, Guadalajara, Monterrey and Puebla get founding points (the Zócalo, the cathedral, the Macroplaza and the Zócalo); the rest get the 1975 built-up peak. The anchor's census numbers are the answer the grid should give.

The second test is WorldPop for 2000 to 2020, on the same cities, centres and zones. GHS-POP is cut to the same twenty years for the comparison.

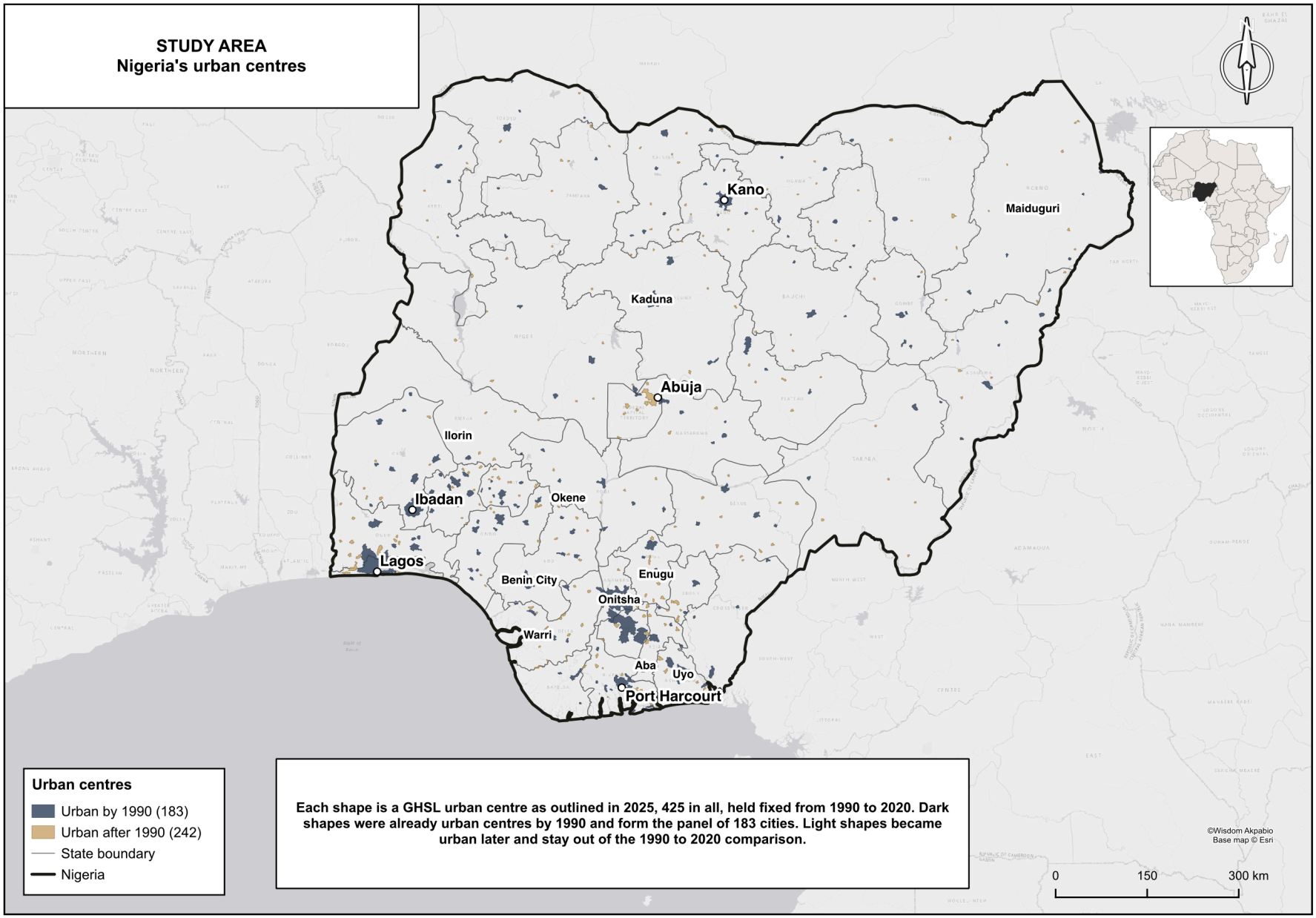


Figure 1. The study area. Each shape is a GHSL urban centre as outlined in 2025. Dark shapes are the 183 that were already urban centres by 1990; light shapes became urban later. Thin lines are state boundaries. Named white dots are the five cities reported one by one.

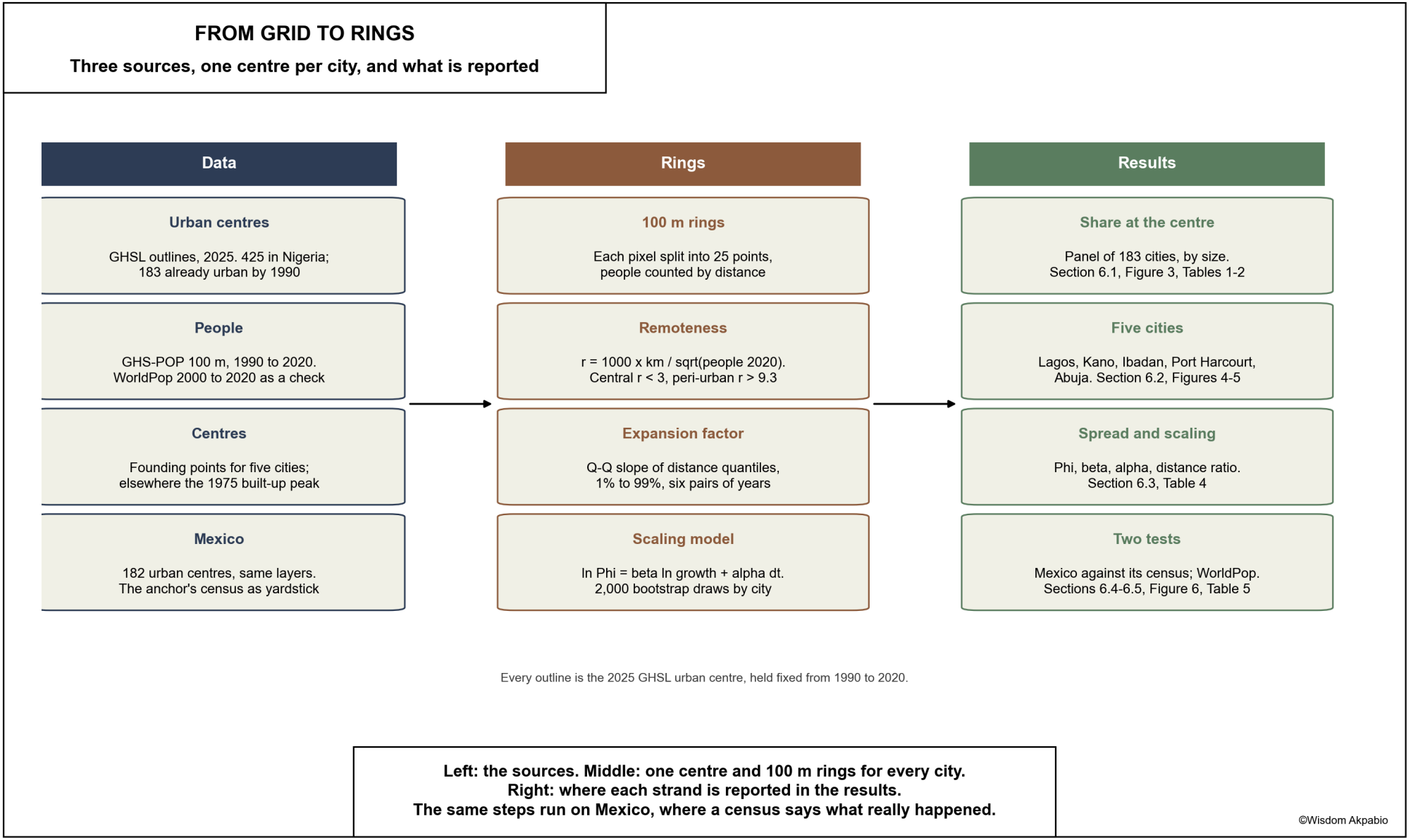


Figure 2. The method. Slate is the data, clay what is done to each city, green what the paper reports and where. Every outline is the 2025 GHSL urban centre, held fixed from 1990 to 2020.

6. Results

6.1 The share at the centre

The central share fell, steadily, in every decade (Table 1). In 1990 **39.1%** of the panel's people lived in a central zone. In 2020 it was **29.9%**. The intermediate zone held its share. The distant and peri-urban zones took the difference.

Table 1. Share of people in each zone, 183 cities, GHS-POP (%)

Zone	1990	2000	2010	2020	Change in people, 1990 to 2020
Central	39.1	37.0	32.0	29.9	+9.74 M
Intermediate	23.0	23.7	24.0	23.8	+11.00 M
Distant	25.2	26.5	29.6	31.3	+16.37 M
Peri-urban	12.7	12.8	14.3	15.0	+7.65 M

The fall is not a few big cities moving the average. The central share fell in **177 of 183** cities, by a median of **13 points** (Figure 3b). Of the 44.8 million people added to the panel, about one in five was added to a central zone, where two in five had lived in 1990.

The exception is size (Table 2). In cities under a million, the median fall was 13 to 14 points. In the fourteen cities over a million it was **3.7 points**. Lagos fell by **1.0**, Port Harcourt by 1.8, Ibadan by 3.7 and Kano by 7.1. Onitsha and Owerri, the two great multi-centred spreads of the south-east, did not move at all. Benin City is the large city that behaved like a small one: its central share fell from 31% to 16%.

Table 2. Change in the central share by city size in 2020, 1990 to 2020

Size in 2020	Cities	Share fell	Median change (points)	Centre lost people
Under 100,000	13	13	−14.0	4
100,000 to 250,000	101	98	−13.9	8
250,000 to 1 million	55	54	−12.7	6
Over 1 million	14	12	−3.7	1

In absolute terms only 19 of the 183 centres lost people. The largest losses were in Akure (128,000 to 97,000), Ikorodu (103,000 to 83,000), Katsina Ala, Yenagoa, Ado on the Karu side of Abuja, Wukari, Dutse, Sagamu and Benin City (273,000 to 266,000). These are mid-sized state capitals and market towns, not the megacities. Section 6.4 explains why the other 164 gains should not be read as growth at the centre.

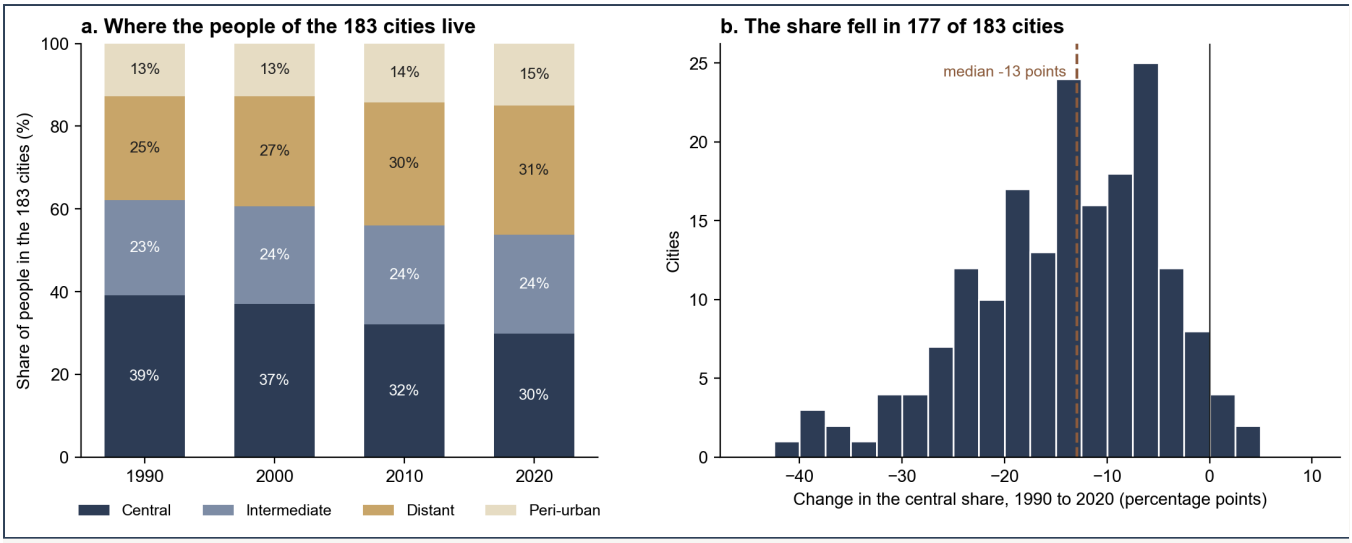


Figure 3. (a) Where the people of the 183 cities lived, by zone, in each year. (b) The change in each city's central share, 1990 to 2020. The dashed line is the median. Only six cities lie to the right of zero.

6.2 Five cities

Table 3. The five named cities, GHS-POP, 1990 to 2020

City	People 1990	People 2020	Central zone reaches	Central 1990	Central 2020	Central share	Φ
Lagos	5.86 M	11.73 M	10.3 km	1.54 M	2.97 M	26.3% to 25.3%	1.04
Kano	2.12 M	4.45 M	6.3 km	1.28 M	2.37 M	60.3% to 53.2%	1.05
Ibadan	1.60 M	3.49 M	5.6 km	0.78 M	1.58 M	48.9% to 45.2%	1.12
Port Harcourt	0.68 M	2.11 M	4.4 km	0.27 M	0.79 M	39.2% to 37.4%	1.06
Abuja	0.18 M	1.13 M	3.2 km	11,000	39,000	6.1% to 3.4%	1.09

Lagos doubled, and its central zone doubled with it. The zone is wide: 10.3 kilometres from Lagos Island takes in Ikoyi, Victoria Island, Apapa, Surulere, Ijesha, Itire and Alaba. Figure 5a shows where the people went. The deep red is not on the island. It is Alaba to the west, Iju and Ojokoro to the north, and the belt through Egbeda, Idimu, Egbe and Ajao Estate. Lagos Island and Ikoyi are pale: GHS-POP gave them few new people, and took almost none away.

Kano still keeps more than half its people within 6.3 kilometres of Gidan Rumfa, the most central of the five. The walled city is red on Figure 5b, with the heaviest gains around Kofar Ruwa, Unguwar Damai and Tudun Yola to the west of the palace and in a broad crescent to the south-west through Rimin Gata, Wagaga and Dorayi to Sabuwar Gandu. There are blue patches by Sharada and Unguwar Sabo, where the model moved people out.

Ibadan has the clearest stretch of the five, Φ = 1.12: its people lived 12% farther from Mapo Hill in 2020 than in 1990, in proportion. Figure 5c shows gains packed into the central circle and a long blue band to its east and south-east, which Section 8 returns to.

Port Harcourt tripled. Its gains on Figure 5d are densest inside the central circle, around the Old GRA and the Township and north of them towards Rumukalagbor, and its northern mainland filled more thinly. The central share barely moved.

Abuja is a different kind of city. Its central zone holds 39,000 people because its origin, the Three Arms Zone, is a district of government buildings at the foot of Aso Rock, outside GHSL's outline of where people live (Figure 5e). The density profile (Figure 4) climbs as it leaves the centre and peaks five kilometres out, in Wuse, Garki and Maitama. A planned capital has a centre of power, not a centre of residence.

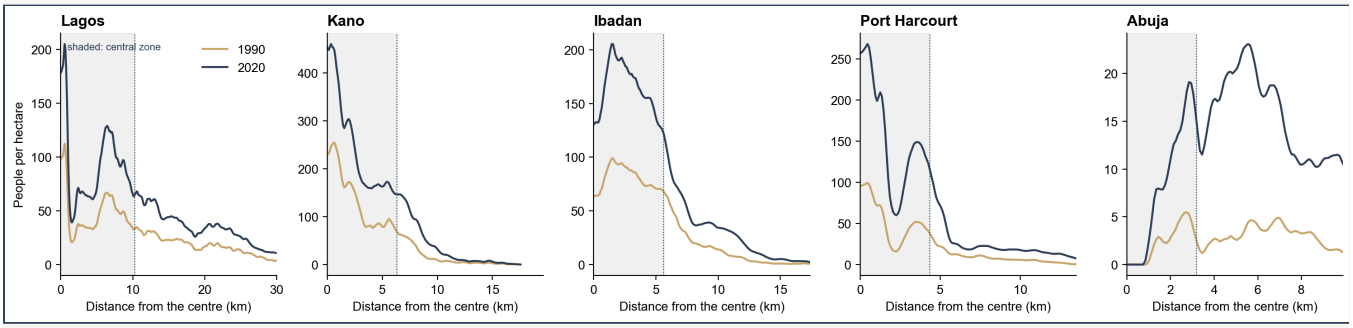


Figure 4. People per hectare by distance from the centre, 1990 (sand) and 2020 (slate), smoothed over 500 metres. The shaded band is the central zone. Lagos's dip near the centre is the lagoon. Abuja's profile rises away from the Three Arms Zone.

6.3 Spread and scaling

The anchor's model says almost nothing in Nigeria (Table 4). β is close to zero, with an interval that includes zero. α is close to the Mexican value. Put together for a panel that grew 2.1 times, the people of 2020 live **18% closer** to the centre than they would in a city that had grown at constant density. Mexico's census says 28% farther.

Table 4. The scaling model. Intervals are 95%, from 2,000 resamples of cities

	Nigeria, GHS-POP, 1990 to 2020	Mexico, GHS-POP, 1990 to 2020	Mexico, census (anchor)
Cities	183	92	69
β	−0.027 (−0.114 to 0.055)	−0.003 (−0.032 to 0.033)	0.60
α	0.0065 (0.0045 to 0.0086)	0.0009 (0.0004 to 0.0014)	0.0057
Mean growth	2.11	1.58	2.16
Distance ratio	0.82	0.81	1.28

Taken at face value, that would be a densifying Nigeria against a spreading Mexico. The middle column says not to take it at face value. On the same grid, Mexico gives a β of zero too. β is a property of the population source before it is a property of the cities.

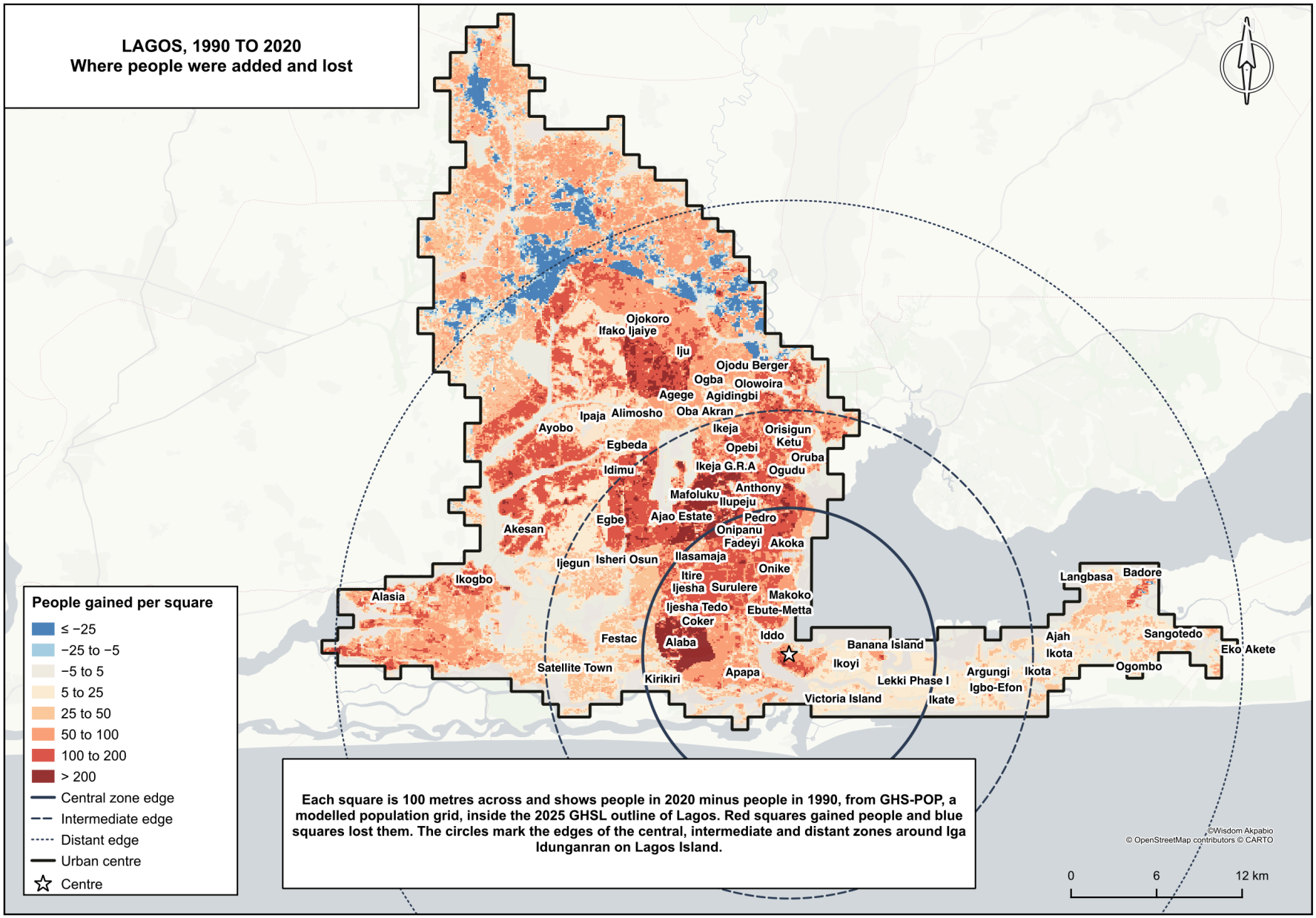


Figure 5a. Lagos. Change in people per 100 metre square, 1990 to 2020, GHS-POP. Circles are the edges of the central (10.3 km), intermediate and distant zones around Iga Idunganran.

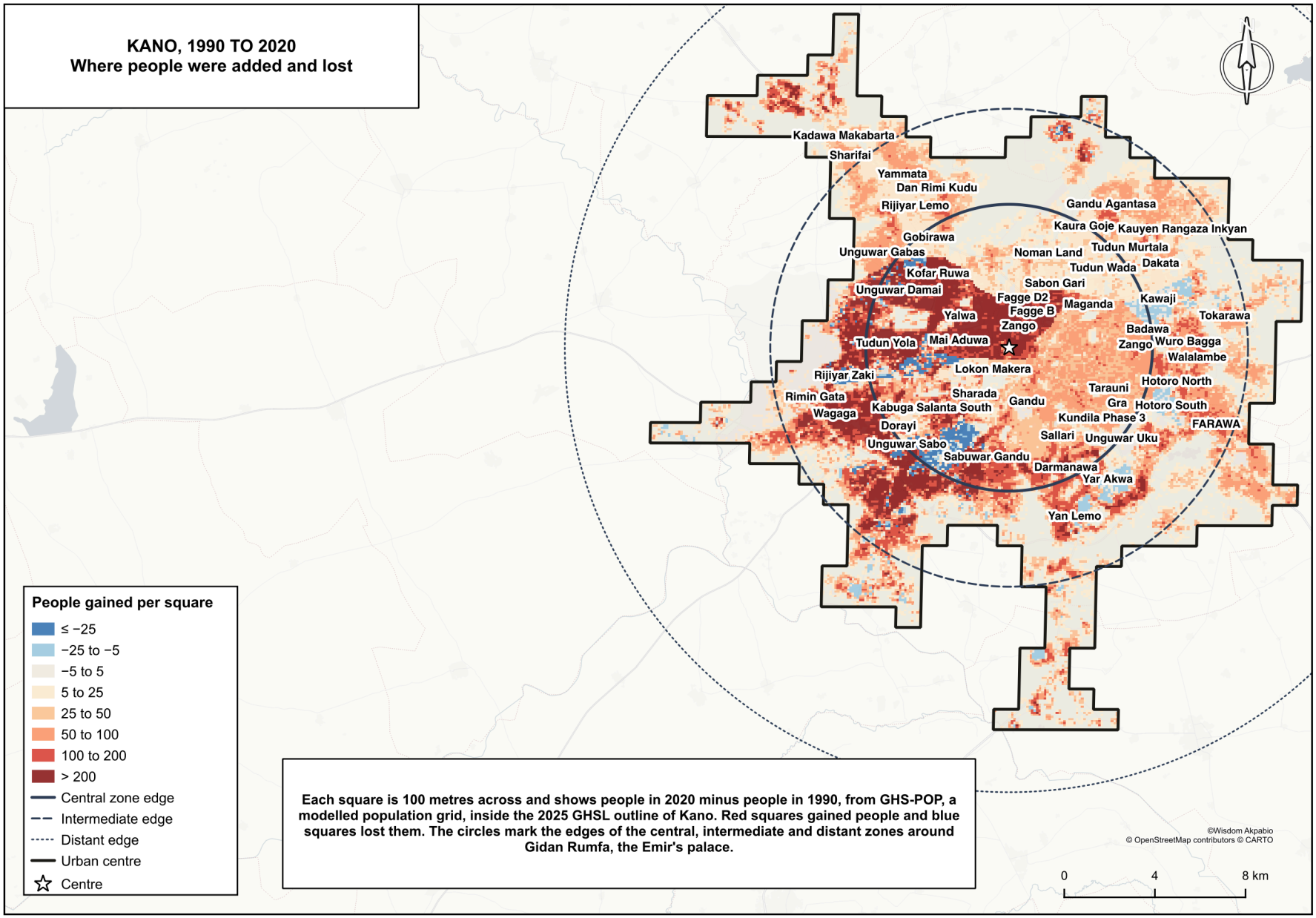


Figure 5b. Kano. Change in people per 100 metre square, 1990 to 2020, GHS-POP. The central zone reaches 6.3 km from Gidan Rumfa.

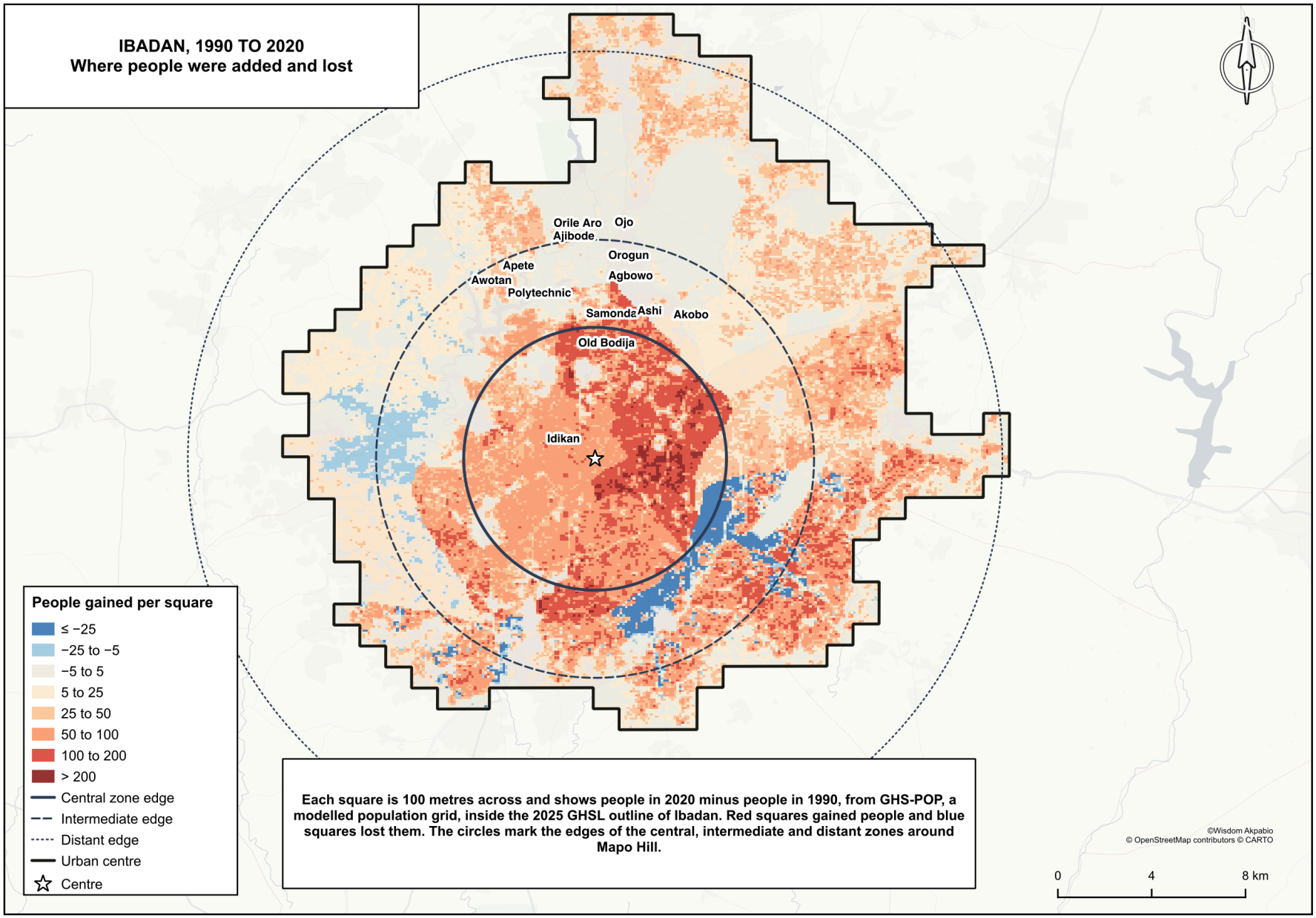


Figure 5c. Ibadan. Change in people per 100 metre square, 1990 to 2020, GHS-POP. The central zone reaches 5.6 km from Mapo Hill. The blue band east of the centre is discussed in Section 8.

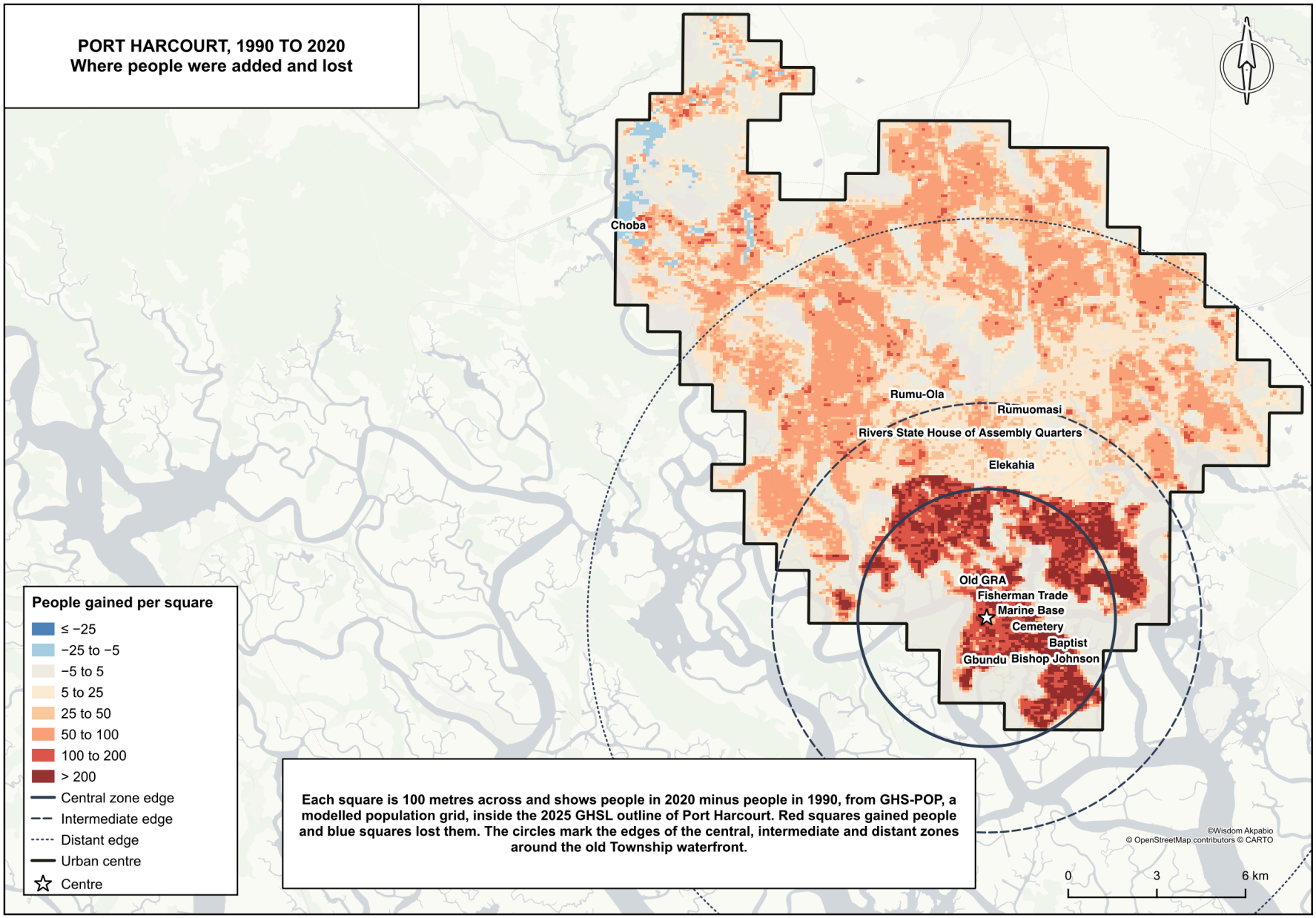


Figure 5d. Port Harcourt. Change in people per 100 metre square, 1990 to 2020, GHS-POP. The central zone reaches 4.4 km from the old Township waterfront.

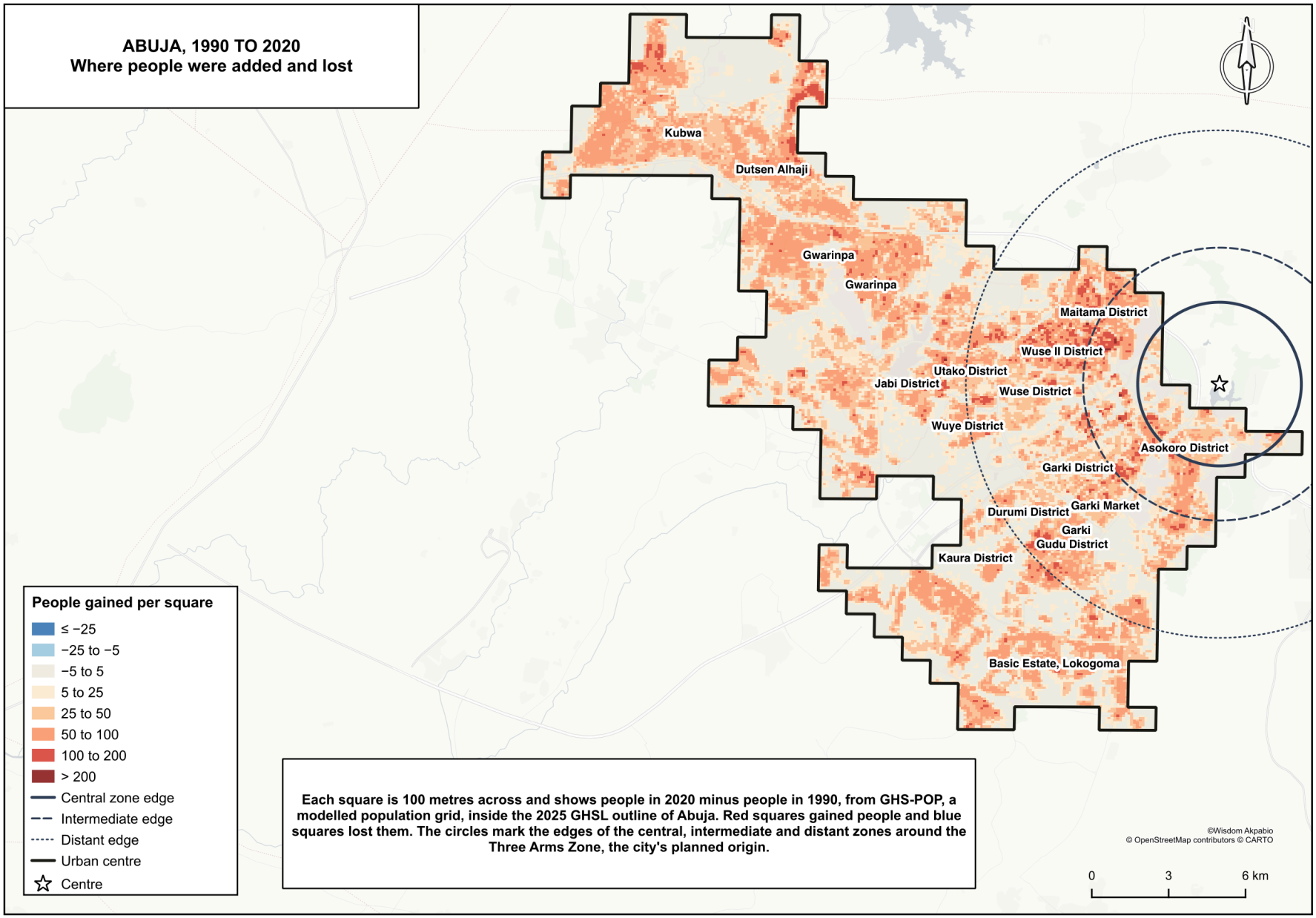


Figure 5e. Abuja. Change in people per 100 metre square, 1990 to 2020, GHS-POP. The Three Arms Zone lies just outside GHSL's outline, so most of the 3.2 km central zone is government ground and hillside.

6.4 The Mexican test

The anchor counted a loss of 2.5 million residents in Mexico's central zones. GHS-POP, through the same steps, finds a **gain of 1.9 million** (Table 5, Figure 6a). Only two of 92 Mexican cities lose central population on the grid: Mexico City (6.09 million to 5.86 million) and Guadalajara (1.03 million to 0.95 million).

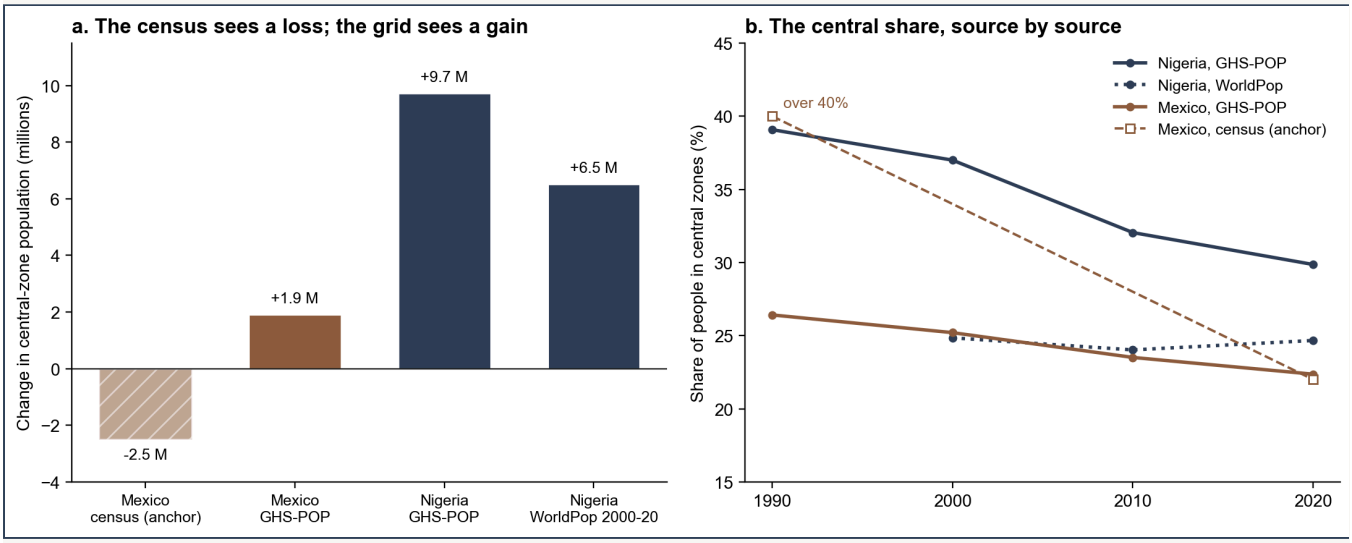


Figure 6. (a) Change in central-zone population, 1990 to 2020 (WorldPop 2000 to 2020). The hatched bar is the Mexican census, from the anchor. (b) The central share by source. The census line joins only its two published end points.

The share behaves better. The census says the central share fell from over 40% to 22%. GHS-POP puts it at 26.4% in 1990 and 22.4% in 2020 (Figure 6b). The end point is right. The start is too low: GHS-POP's 1990 Mexico is already spread out, and the fall from it is too small.

So the grid can be trusted for the direction of the share and not for the count. A fall in the share on GHS-POP is, if anything, smaller than the true fall. A gain at the centre on GHS-POP could be a loss on the ground.

Table 5. The two tests

	Central change	Central share	Cities where the share fell	Centres that lost people
Mexico, census, 1990 to 2020 (anchor)	−2.5 M	over 40% to 22%	not reported	most large cities
Mexico, GHS-POP, 1990 to 2020	+1.91 M	26.4% to 22.4%	88 of 92	2 of 92
Nigeria, GHS-POP, 1990 to 2020	+9.74 M	39.1% to 29.9%	177 of 183	19 of 183
Nigeria, GHS-POP, 2000 to 2020	+6.22 M	37.0% to 29.9%	178 of 183	28 of 183
Nigeria, WorldPop, 2000 to 2020	+6.52 M	24.9% to 24.7%	59 of 183	0 of 183

6.5 A second grid

WorldPop agrees that the central zones gained about six and a half million people between 2000 and 2020. It does not agree on anything else. Its central share is flat. No centre loses people. Its Φ for the five named cities is between 1.01 and 1.03, and its β is slightly negative. Its map of each city barely changes shape between 2000 and 2020; it looks like one pattern scaled up to each year's total.

That makes WorldPop a weak check on spread, in either direction. What it does rule out is that the central gain is a quirk of GHS-POP alone. Both grids put more people in Nigeria's central zones in 2020 than in 2000. Neither grid, as the Mexican test shows, would be able to show it if the opposite were true.

7. Discussion

The Mexican paper found a doughnut: a centre that lost people while the ring around it filled. This paper cannot find or rule out a Nigerian doughnut, and it says so plainly. It does find a centre that holds less. Between 1990 and 2020 the central zones of Nigerian cities took about one in five of the people added to those cities, little more than half their 1990 share. In 177 of the 183 cities, the centre's share fell. That much survives every test here, and the Mexican test says it is more likely too small than too large.

The size result is the part that would have been missed with five cities. Lagos, Kano, Ibadan and Port Harcourt held their central shares far better than the median town. One reason is arithmetic: a zone that scales with the square root of population gives Lagos a central circle of 10 kilometres, which includes ground that was suburb in 1990. Another is that big cities fill in before they spread out, with storeys on old compounds and every gap on the mainland built. The small towns, where land at the edge is closer and cheaper and nothing at the centre gets taller, spread instead. Akure, Ado Ekiti, Yenagoa and Wukari each lost between 25 and 40 points of their central share. That pattern is consistent with what GHS-POP is good at seeing, new building at the edge, so it is the part of the result most worth taking to planners.

The absolute numbers are another matter. A gain of 9.7 million people in central zones would, if true, mean Nigeria's old cores became much denser while Mexico's emptied. Perhaps they did. Nigeria's natural growth is far higher than Mexico's was, and the old cores of Kano and Ibadan are full of family compounds that get more crowded, not less. But the grid that says so turned Mexico's emptying cores into filling ones. The honest reading is that the number cannot be known from these sources.

The scaling model teaches the same lesson in a different way. The anchor's β of 0.60 is a statement about cities only if the population data can see people move. On GHS-POP, Mexico's β drops to zero, and so does Nigeria's. A cross-country comparison of β built on global grids would compare grids, not cities.

8. Limitations

No census to check against. Nigeria's censuses were in 1991 and 2006. The one planned for 2023 was postponed. GHS-POP and WorldPop are both built on projected totals for local government areas, and the projection inside each area is the model's. There is no Nigerian equivalent of the Mexican

block count.

Built volume is not residence. GHS-POP spreads people over built-up volume. Lagos Island's markets, Kano's Kurmi market and the Dugbe commercial district in Ibadan are dense buildings with few residents at night. The grid keeps people in them. That is the likely reason it misses central loss in Mexico, and it would hide the same in Nigeria.

Lagos's northern arm. The 2025 outline crosses into Ogun State. Large blue patches there on Figure 5a are model losses at the very edge of the city, where building surely grew. They are a reminder that a 100 metre change on GHS-POP is not an observation.

Ibadan's eastern band. A long, straight-edged band of losses runs east and south-east of Mapo Hill. No account of the city we know of explains it. It is the clearest sign on any plate that the model can move people between epochs for reasons of its own, and it sits partly inside the central and intermediate zones.

Abuja's centre is not a residential centre. The Three Arms Zone lies outside GHSL's outline. Abuja is reported as the exception, and it is outside the panel because GHSL dates its urban centre to 1995.

Water in the rings. The central zones of Lagos and Port Harcourt are partly lagoon, sea and creek. Ring areas include the water, as in the anchor, so their density levels are low; changes are unaffected.

One centre for a spread. Onitsha (4.9 million) and Owerri (3.2 million) are continuous settled regions with many old towns. A single 1975 peak does not describe them. Their central shares did not move, and they are in the totals.

A fixed outline. Every ring is inside the 2025 outline in every year. In 1990 much of the peri-urban zone of most cities was farmland. That is the design, but it means the 1990 denominators include land with almost nobody on it.

9. Policy

National Population Commission. When the next census is held, publish counts by enumeration area with boundaries, and a crosswalk to 1991 and 2006 for the old-core local government areas: Lagos Island, Kano Municipal, Ibadan North and Ibadan South-East, Port Harcourt and Oredo in Benin City. Those few tables would allow the Mexican count to be run in Nigeria and settle the question this paper cannot.

Lagos State Ministry of Physical Planning and Urban Development, and the Kano State Urban Planning and Development Authority. Do not use a satellite population grid to argue that Lagos Island, Isale Eko, or Kano's walled city is filling up or emptying out. Commission a household count in those places before regeneration or relocation plans that assume either. The grid will show people living in the market.

Ondo, Ekiti, Bayelsa and Taraba State governments. The fastest relative shift outward in the panel is in their capitals and market towns: Akure's central share fell from 41% to 15%, Ado Ekiti's from 51% to 22%, Yenagoa's from 40% to 12%, and Wukari's from 70% to 30%. New schools, clinics and roads planned around the old towns will sit where a shrinking share of people live. Plan the next ones for the distant and peri-urban zones, where most of the growth went.

Researchers using global grids. Report the share by zone, which survives the Mexican test, alongside any count. Do not compare β or α across countries on GHS-POP or WorldPop and read the difference as a difference between cities.

10. Conclusions

Did the centre of Nigerian cities hold less of their people in 2020 than in 1990? Yes. The central share fell from 39.1% to 29.9% across 183 cities and fell in 177 of them. The towns did most of the falling, by a median of 13 points. The cities over a million did least: Lagos's central share barely moved, Kano still keeps more than half its people within 6.3 kilometres of the Emir's palace, and Abuja's centre was never a place to live.

Did the centre lose people, as in Mexico? That cannot be answered from these sources, and the paper is built to show why. The same grid that puts 9.7 million more people in Nigeria's central zones puts 1.9 million more in Mexico's, where the census counted 2.5 million fewer. The second grid hardly changes its shape at all.

So the defensible claim is about where growth went, not about who left. Most of Nigeria's added urban population settled away from the old centres, and in the small and middle-sized cities the move was large. What happened inside the old centres, in Lagos Island, in Kano's walls, under Mapo Hill, needs a count of people, house by house. Until the census returns, the satellite can tell us that the city spread. It cannot tell us whether its heart emptied.

Data availability

The write-up, the tables, the figures and the code are in this repository. GHSL, WorldPop, geoBoundaries, Natural Earth and OpenStreetMap keep their own licences. Rasters are not committed; `src/download_ghsl.py` fetches them again.

Write-up	`notes/results.md`, `notes/metric_trace.md`
Tables	`data/processed/summary*.csv`, `zone_shares.csv`, `share_by_size.csv`, `city_central_share.csv`, `central_losers.csv`, `named_cities*.csv`, `expansion_factors*.csv`
Rings	`data/processed/rings.csv.gz`, `rings_worldpop.csv.gz`, `rings_mx.csv.gz`
Maps	`maps/nigeria_study_area.png`, `maps/{city}_change.png`, `maps/centre_check_*.png`
Charts	`charts/method_flowchart.png`, `zone_shares.png`, `density_profiles.png`, `tests.png`

Code	`src/download_ghsl.py`, `src/radial.py`, `src/analyse.py`, `src/tables.py`, `src/charts.py`, `src/flowchart.py`, `src/plate_data.py`, `src/check_centres.py`, `qgis/build_plates.py`

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